DESCRIPTION OF COURSEWORK

Course Code	EEE211*	
Course Name	Data Structure	
Lecturer	Nowshath Kadhar Batcha	
Academic Session	2026/04	
Assessment Title	Assignment	

A. Introduction/ Situation/ Background Information

This assignment is to evaluate the ability of students in applying the concepts of binary search, recursion, and sorting problems.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 To describe the concepts of data structures (such as stack, queue and etc.) , logical structure and storage structure of data and their basic operation.

CLO 2 To demonstrate ability to solve practical problems using suitable data structures and methods.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

You are required to **answer all the assignment questions** in Section F. This is an individual assignment. You are required to **prepare one hardcopy and one softcopy of a typewritten report detailing the answers to the assignment questions**. The deadline of this assignment is on **22 Jun 2026.** Late submission will result in a penalty. The report should include the following sections in the same sequence,

- **Cover page:** student name and student id
    - Assignment Cover Pages (must be filled and signed by student) - *attached*
- **C++ programs section:** source code with comments
    - Include C++ program for each task from Section F
- **Results section:** detailed description of results
    - Include all programs results/outputs with descriptions in your own words

Marking rubrics: Include at the end of report/document – attached.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Linked List	25
2.	Stack	25
3.	Queue	25
4.	Linear Search	25
	TOTAL	**100**

F. Task(s)

Section A: Linked List (25%)

A music streaming app maintains a **circular playlist** of song IDs. Since the playlist repeats continuously, the last song links back to the first song. The app should allow the user to add new songs at different positions in the playlist or remove songs when needed. After every modification, the updated circular playlist must be displayed.

Given the scenario, accomplish the following tasks using C++

- Create a circular linked list that stores integers.
- Allow the user to:
    - Insert nodes at the beginning, end, and specified position.
    - Delete nodes from the beginning, end, and specified position.
- After each operation, display the circular linked list.

[25 marks]

Section B: Stack (25%)

A browser keeps track of recently visited pages using a **stack**. Each new page visited is pushed onto the stack, and when the user clicks the back button, the most recent page is removed first. Since memory is limited, the browser only stores up to **10 pages**.

Write a C++ program to:

- Implement a stack using an array of size 10.
- Provide push, pop, and display operations.
- Handle overflow and underflow conditions.

[25 marks]

Section C: Queue (25%)

A supermarket has two billing counters, and each counter stores customer token numbers in a **circular queue**. At the end of the day, both queues need to be merged into a single queue for reporting purposes. During merging, all customers with **even token numbers** must be placed first, followed by all customers with **odd token numbers**. No sorting algorithm is allowed; the arrangement must be achieved only during the merging process.

Write a complete C++ program to do the following:

- Create two circular queues (queue1 and queue2) using linked lists.
- Allow the user to input positive integers into both queues.
- Merge both queues into a third circular queue (mergedQueue) so that.
    - All even numbers appear first (left side).
    - All odd numbers appear after (right side).

Example:

queue1 → 4, 7, 2

queue2 → 9, 6, 5

mergedQueue → 4, 2, 6, 7, 9, 5

You are not allowed to use any sorting algorithm in this task — only the merging process should determine the order.

[25 marks]

Section D: Linear Search (25%)

A school stores the names of 10 students who registered for a workshop in an array. When a teacher enters a student's name, the system must check whether that student is registered and display the position of the name if found.

Write a C++ program to:

- Create an array of 10 strings.
- Ask the user to input a name to search.
- Perform a linear search to check if the name exists.
- Print the result (found/not found, and index if found).

MARKING RUBRICS

Component Title	Linked List			Percentage (%)		25	
Criteria	Score and Descriptors					Weight (%)	Marks
	(20-25)	(15-20)	(10-15)	(5-10)	(0-5)		
C++ program and Results	Code free of errors. Excellent use of variables and organization. Output provided	No errors in the coding able to compile the code. Not efficient use of variable, output not included	Some mistakes in the coding Not efficient use of variable, missing some tasks	Few mistakes in the coding Not efficient use of variable, missing few tasks	Many mistakes in the coding unable to compile the code, missing many tasks	25	
					SUBTOTAL	25	

Component Title	Stack			Percentage (%)		25	
Criteria	Score and Descriptors					Weight (%)	Marks
	(20-25)	(15-20)	(10-15)	(5-10)	(0-5)		
C++ program and Results	Code free of errors. Excellent use of variables and organization. Output provided	No errors in the coding able to compile the code. Not efficient use of variable, output not included	Some mistakes in the coding Not efficient use of variable, missing some tasks	Few mistakes in the coding Not efficient use of variable, missing few tasks	Many mistakes in the coding unable to compile the code, missing many tasks	25	
					SUBTOTAL	25	

Component Title	Queue			Percentage (%)		25	
Criteria	Score and Descriptors					Weight (%)	Marks
	(20-25)	(15-20)	(10-15)	(5-10)	(0-5)		
C++ program and Results	Code free of errors. Excellent use of variables and organization. Output provided	No errors in the coding able to compile the code. Not efficient use of variable, output not included	Some mistakes in the coding Not efficient use of variable, missing some tasks	Few mistakes in the coding Not efficient use of variable, missing few tasks	Many mistakes in the coding unable to compile the code, missing many tasks	25	

					SUBTOTAL	**25**	
Component Title	Linear Search			**Percentage (%)**		**25**	
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	(20-25)	**(15-20)**	**(10-15)**	**(5-10)**	**(0-5)**		
C++ program and Results	**Code free of errors. Excellent use of variables and organization. Output provided**	**No errors in the coding able to compile the code. Not efficient use of variable, output not included**	**Some mistakes in the coding Not efficient use of variable, missing some tasks**	**Few mistakes in the coding Not efficient use of variable, missing few tasks**	**Many mistakes in the coding unable to compile the code, missing many tasks**	**25**	
						SUBTOTAL	
						TOTAL	

Note to students: Please print out and attach this appendix together with the submission of coursework.